

Projectile System

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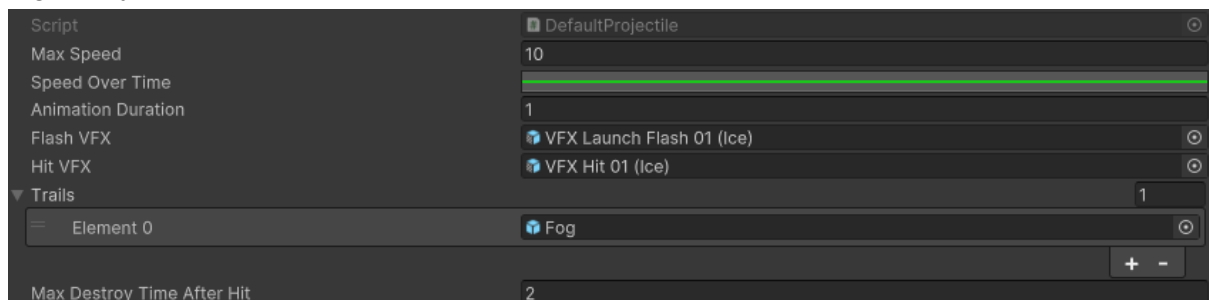
Overview

This module includes a simple projectile system that allows you to shoot projectiles. The system automatically handles the spawning of launch flash effects and hit effects.

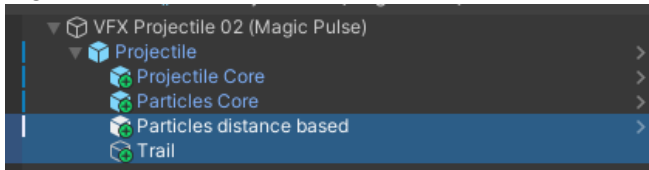
All included modules of an asset including the according documentation can be found under: *SF Studio* → *Shared* → *Modules*

Default Projectile

Each projectile has the Default Projectile script attached. This script requires a Collider and a Rigidbody component.



You can configure the following properties:

Max Speed	Defines the maximum speed of the projectile.
Speed over Time	Provides fine grained control over the projectile speed using a curve. For example, you can make the projectile start fast and slow down toward the end of its lifetime.
Animation Duration	Defines the duration in seconds for the speed curve animation.
Flash VFX	Launch flash effect that is spawned when the projectile is instantiated.
Hit VFX	Impact effect that is spawned when the projectile collides with another object.
Trails	A list of distance based child particles or trail objects that should not be destroyed immediately on impact. These objects will be detached from the projectile so they can continue playing even after the projectile itself is destroyed. This helps achieve smoother visual transitions. E.g. 
Max Destroy Time	Defines the delay before the projectile is destroyed after impact. This is typically based on the particle system duration. For non-particle based components such as Trail Renderers, this value will be used instead.

Projectile Spawner

The Projectile Spawner allows you to select from a list of different projectiles and spawn them simply by pressing the left mouse button.

You can configure the following properties:

General	
Fire Point	Transform reference that defines the position and direction from which projectiles are spawned.
Default Target	Target object the projectiles will travel toward.
Fire Rate	Defines how many projectiles can be fired per second.
Targeting Mode	
Mouse Targeting	If enabled, the active target that the projectile is fired at is determined by the current mouse position.
Mouse Target Layers	Defines which layers are considered when calculating the mouse based target position.
Effect & Gun Control	
Effect Name	TextMeshPro UGUI reference that displays the name of the currently selected projectile.
Next/ Previous Key	Customizable input keys used to switch between available projectiles.
Gun	Optional visual object representing the firing source, such as a pistol or machine gun. This object will automatically rotate to face the target.
VFX Overlay Settings	
Enable VFX Overlay	Enables the optional VFX layer overlay system. When enabled, following will happen: On impact, the hit VFX will be moved to the specified overlay layer and rendered by the overlay camera. This ensures the effect is always visible and not partially clipped by surrounding geometry, such as walls.
VFX Layer Overlay	Defines the layer used for rendering overlay VFX.
Overlay Cam	Separate camera that renders only the specified VFX overlay layer.
Projectiles	List of available projectile prefabs that can be selected and spawned.

